

Online Resource 1

Hydrothermal carbonization-assisted dewatering of cattle manure digestate: compression mechanisms and phosphorus and potassium partitioning for integrated valorization

Waste and Biomass Valorization

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Contents

- Table S1 Central composite design matrix and measured responses
- Table S2 Regression coefficients and model statistics for the response surface models
- Table S3 Step-end piston-press data for all compression tests
- Table S4 Point EDS compositions (at.%)
- Table S5 TOC/TN analysis sequence
- Fig. S1 Response surface contour plots
- Fig. S2 Responses as functions of the severity factor
- Fig. S3 Backscattered-electron micrographs
- Fig. S4 Secondary-electron micrographs at $\times 250$
- Fig. S5 TOC and TN against position in the analytical sequence
- Fig. S6 Full-range FTIR spectra of individual samples

Table S1 Central composite design matrix with measured responses of the HTC products. Moisture is not listed because it was measured only on pressed cakes (manuscript Table 4). Run 21 was not analysed. Runs 13 and 27 were logged at 10.4% TS; the design level of 18.41% was used (manuscript Section 2.2). K values marked * exceeded the calibration range

Run	T (°C)	t (min)	TS (%)	log R0	Ash (%)	VS (%)	pH	EC (mS/cm)	TOC (mg/L)	TN (mg/L)	K (mg/L)	PO4 (mg/L)
1	160	60	5	3.54	18.81	81.19	6.63	4.37	7937	2252	569	462 ± 0.8
2	260	60	5	6.49	18.63	81.37	6.39	3.90	7846	2273	536	229 ± 7.9
3	160	120	5	3.85	18.63	81.37	6.81	4.52	7516	2193	643	544 ± 3.8
4	260	120	5	6.79	27.72	72.28	6.32	4.65	11007	3045	660	254 ± 0.2
5	160	60	15	3.54	19.61	80.39	7.30	7.99	2058	820	1317	341 ± 3.6
6	260	60	15	6.49	27.45	72.55	6.56	9.62	4088	1300	1939	943 ± 0.3
7	160	120	15	3.85	23.76	76.24	6.69	8.66	4649	1351	1749	657 ± 2.4
8	260	120	15	6.79	27.18	72.82	6.49	9.95	2963	1157	1948	665 ± 1.6
9	294.1	90	10	7.67	29.13	70.87	6.77	8.06	4469	1270	1313	496 ± 0.7
10	125.9	90	10	2.72	21.36	78.64	7.14	7.07	7866	2443	1030	288 ± 9.9
11	210	140.46	10	5.39	22.33	77.67	6.48	8.29	10404	2710	1279	922 ± 1.8
12	210	39.54	10	4.84	20.00	80.00	6.31	8.33	10063	2627	1243	1249 ± 0.7
13	210	90	18.41	5.19	22.77	77.23	6.22	11.71	7855	2244	2216*	1460 ± 1.7
14	210	90	1.59	5.19	15.84	84.16	6.21	2.20	3335	1364	231	254 ± 0.6
15	160	60	5	3.54	18.81	81.19	6.58	4.63	7722	2223	635	633 ± 0.6
16	260	60	5	6.49	26.73	73.27	6.80	4.94	7571	2076	709	475 ± 2.4
17	160	120	5	3.85	20.79	79.21	6.95	4.38	11417	3167	599	528 ± 1.6
18	260	120	5	6.79	28.00	72.00	6.90	4.70	1802	792	704	387 ± 3.4
19	160	60	15	3.54	20.00	80.00	6.85	9.01	7471	2156	1725	253 ± 3.0
20	260	60	15	6.49	23.76	76.24	6.43	10.09	7523	2141	2023*	1059 ± 2.2
21	160	120	15	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
22	260	120	15	6.79	28.43	71.57	6.54	9.90	878	930	1948	844 ± 1.8
23	294.1	90	10	7.67	28.71	71.29	6.69	8.30	927	952	1393	494 ± 0.4
24	125.9	90	10	2.72	18.45	81.55	7.71	5.46	956	942	1003	125 ± 0.5
25	210	140.46	10	5.39	22.33	77.67	6.38	8.23	885	826	1400	917 ± 3.0
26	210	39.54	10	4.84	21.15	78.85	6.39	8.38	791	826	1314	1353 ± 0.2
27	210	90	18.41	5.19	21.00	79.00	6.11	12.14	673	651	2269*	1574 ± 45.8
28	210	90	1.59	5.19	18.81	81.19	6.69	2.41	466	584	229	393 ± 0.2
29	210	90	10	5.19	22.33	77.67	6.39	8.03	362	521	1289	999 ± 0.1
30	210	90	10	5.19	20.79	79.21	6.37	8.16	929	1216	1358	882 ± 4.9

Table S2 Coefficients of the second-order response surface models in coded variables (T: (T – 210)/50; t: (t – 90)/30; TS: (TS – 10)/5), with standard errors (SE), t statistics and p values. For each response, the "Model" row gives R², adjusted R², the F statistic and its p value. n = 29 runs

Response	Term	Coefficient / R ²	SE / adj. R ²	t / F	p
Ash (%)	β_0	21.32	1.45	14.7	<0.001
Ash (%)	T	2.797	0.4052	6.9	<0.001
Ash (%)	t	1.026	0.4052	2.53	0.020
Ash (%)	TS	1.055	0.4052	2.6	0.017
Ash (%)	T ²	1.345	0.594	2.26	0.035
Ash (%)	t ²	0.2985	0.594	0.503	0.621
Ash (%)	TS ²	-0.3546	0.594	-0.597	0.558
Ash (%)	T×t	0.4621	0.5391	0.857	0.402
Ash (%)	T×TS	-0.1254	0.5391	-0.233	0.818
Ash (%)	t×TS	-0.1371	0.5391	-0.254	0.802
Ash (%)	Model	0.8013	0.7072	8.51	<0.001
pH	β_0	6.364	0.1548	41.1	<0.001
pH	T	-0.1646	0.04327	-3.8	0.001
pH	t	0.007567	0.04327	0.175	0.863
pH	TS	-0.02567	0.04327	-0.593	0.560
pH	T ²	0.2689	0.06344	4.24	<0.001
pH	t ²	0.02585	0.06344	0.407	0.688
pH	TS ²	-0.00331	0.06344	-0.0522	0.959
pH	T×t	0.01265	0.05758	0.22	0.828
pH	T×TS	-0.06485	0.05758	-1.13	0.274
pH	t×TS	-0.0764	0.05758	-1.33	0.200
pH	Model	0.7293	0.6011	5.69	<0.001
EC (mS/cm)	β_0	8.213	0.4333	19	<0.001
EC (mS/cm)	T	0.4226	0.1211	3.49	0.002
EC (mS/cm)	t	0.03578	0.1211	0.295	0.771
EC (mS/cm)	TS	2.584	0.1211	21.3	<0.001
EC (mS/cm)	T ²	-0.4716	0.1775	-2.66	0.016
EC (mS/cm)	t ²	-0.08812	0.1775	-0.496	0.625
EC (mS/cm)	TS ²	-0.5096	0.1775	-2.87	0.010
EC (mS/cm)	T×t	0.0001818	0.1611	0.00113	0.999
EC (mS/cm)	T×TS	0.2827	0.1611	1.75	0.095
EC (mS/cm)	t×TS	0.02982	0.1611	0.185	0.855
EC (mS/cm)	Model	0.9635	0.9462	55.8	<0.001
K (mg/L)	β_0	1328	62.72	21.2	<0.001
K (mg/L)	T	97.34	17.53	5.55	<0.001
K (mg/L)	t	25.71	17.53	1.47	0.159
K (mg/L)	TS	588.1	17.53	33.6	<0.001
K (mg/L)	T ²	-55.76	25.7	-2.17	0.043
K (mg/L)	t ²	-11.84	25.7	-0.461	0.650
K (mg/L)	TS ²	-37.68	25.7	-1.47	0.159
K (mg/L)	T×t	-24.5	23.32	-1.05	0.307
K (mg/L)	T×TS	75.24	23.32	3.23	0.004
K (mg/L)	t×TS	11.54	23.32	0.495	0.627
K (mg/L)	Model	0.9846	0.9773	135	<0.001
PO4 (mg/L)	β_0	982.9	161.8	6.07	<0.001
PO4 (mg/L)	T	61.53	45.22	1.36	0.190
PO4 (mg/L)	t	-39.18	45.22	-0.866	0.397
PO4 (mg/L)	TS	219.4	45.22	4.85	<0.001
PO4 (mg/L)	T ²	-267.3	66.29	-4.03	<0.001
PO4 (mg/L)	t ²	1.152	66.29	0.0174	0.986
PO4 (mg/L)	TS ²	-66.1	66.29	-0.997	0.331
PO4 (mg/L)	T×t	-82.81	60.16	-1.38	0.185
PO4 (mg/L)	T×TS	147.1	60.16	2.44	0.024
PO4 (mg/L)	t×TS	23.98	60.16	0.399	0.695
PO4 (mg/L)	Model	0.7702	0.6614	7.08	<0.001
TOC (mg/L)	β_0	293.4	2722	0.108	0.915
TOC (mg/L)	T	-588.9	760.7	-0.774	0.448
TOC (mg/L)	t	-220.4	760.7	-0.29	0.775
TOC (mg/L)	TS	-732.4	760.7	-0.963	0.348
TOC (mg/L)	T ²	1517	1115	1.36	0.190
TOC (mg/L)	t ²	2217	1115	1.99	0.061

Response	Term	Coefficient / R ²	SE / adj. R ²	t / F	p
TOC (mg/L)	TS ²	1350	1115	1.21	0.241
TOC (mg/L)	T×t	-875.3	1012	-0.865	0.398
TOC (mg/L)	T×TS	150.6	1012	0.149	0.883
TOC (mg/L)	t×TS	-505.2	1012	-0.499	0.623
TOC (mg/L)	Model	0.264	-0.08466	0.757	0.655
TN (mg/L)	β0	800.6	594.3	1.35	0.194
TN (mg/L)	T	-147.6	166.1	-0.889	0.385
TN (mg/L)	t	-30.73	166.1	-0.185	0.855
TN (mg/L)	TS	-181	166.1	-1.09	0.290
TN (mg/L)	T ²	282.6	243.5	1.16	0.260
TN (mg/L)	t ²	404.8	243.5	1.66	0.113
TN (mg/L)	TS ²	215.2	243.5	0.884	0.388
TN (mg/L)	T×t	-172.1	221	-0.779	0.446
TN (mg/L)	T×TS	76.36	221	0.345	0.734
TN (mg/L)	t×TS	-107.8	221	-0.488	0.631
TN (mg/L)	Model	0.2417	-0.1175	0.673	0.724

Table S3 Step-end data of the piston-press tests: piston position reading (mm) and cumulative filtrate (g) at the end of each pressure step. The continuous displacement record was not recoverable; these are the measured values used in the analysis

Sample	TS (%)	Step	Pressure (bar)	Time (min)	Piston reading (mm)	Cumulative filtrate (g)
RD 1	15	1	50	10	510.5	98.73
		2	100	20	516.8	135.34
		3	150	30	520.7	157.17
		4	200	45	525.1	175.82
RD 2	15	1	50	10	508.3	91.55
		2	100	20	515.6	134.77
		3	150	30	521.5	172.5
		4	200	45	528.2	202.46
RD 3	15	1	50	10	506.4	88.94
		2	100	20	513.1	122.52
		3	150	30	517.3	145.55
		4	200	45	522	174.43
RD 4	10	1	50	10	489.2	112.83
		2	100	20	510.7	162.51
		3	150	30	520.3	196.14
		4	200	45	528.3	232.77
RD 5	10	1	50	10	510.3	99.13
		2	100	20	518.6	136.41
		3	150	30	523.9	165.65
		4	200	45	530.9	217.74
RD 6	5	1	50	10	476.4	71.28
		2	100	20	501.6	98.17
		3	150	30	516.8	162.37
		4	200	45	532.5	254.31
RD 7	5	1	50	10	469.8	59.66
		2	100	20	494.8	86.29
		3	150	30	505	106.28
		4	200	45	513.9	138.78
RD 8	5	1	50	10	468.5	75.75
		2	100	20	494.6	96.61
		3	150	30	504.6	111.83
		4	200	45	520.1	136.61
RD 9	1.59	1	50	10	583.5	535.55
		2	100	20	586.3	542.34
		3	150	30	587	544.2
		4	200	45	587.7	547.73
HTC 1	15	1	50	10	480	191.91
		2	100	20	537.5	259.95
		3	150	30	549.6	325.25
		4	200	45	560.5	357.87
HTC 2	15	1	50	10	572.9	519.3
		2	100	20	576.5	535.92
		3	150	30	578.3	544.21
		4	200	45	580.1	552.13
HTC 3	15	1	50	10	573.2	534.68
		2	100	20	576.9	553.24
		3	150	30	578.6	561.42
		4	200	45	580.5	570.17
HTC 4	10	1	50	10	459.5	34.45
		2	100	20	487.5	44.91
		3	150	30	497.6	52.91
		4	200	45	504.8	61.97
HTC 5	10	1	50	10	577.3	490.01
		2	100	20	580	505.67
		3	150	30	581.3	511.17
		4	200	45	582.5	517.09
HTC 6	5	1	50	10	583.9	452.92
		2	100	20	585.2	457.54
		3	150	30	586	460.09
		4	200	45	586.9	462.4
HTC 7	5	1	50	10	501	73.88
		2	100	20	513.3	98

Sample	TS (%)	Step	Pressure (bar)	Time (min)	Piston reading (mm)	Cumulative filtrate (g)
HTC 8	5	3	150	30	519.2	115.13
		4	200	45	524.2	172.695
		1	50	10	583.4	543.79
		2	100	20	584.8	549.23
		3	150	30	585.6	551.68
HTC 9	1.59	4	200	45	586.5	554.29
		1	50	10	586.1	559.52
		2	100	20	587	564.82
		3	150	30	587.6	566.49
		4	200	45	588.3	568.28

Table S4 Point EDS compositions (atomic %) of the four characterized cakes. Blank elements were not detected (0.0). The point flagged "metal particle" was excluded from all analyses

Sample	Point	C	O	Na	Mg	Si	P	S	Cl	K	Ca	Fe	Note
HTC 4	S1(1)_pt1	41.43	41.15	0.00	0.17	16.77	0.14	0.00	0.03	0.13	0.19	0.00	
HTC 4	S1(1)_pt2	48.75	33.84	0.00	0.14	16.68	0.22	0.00	0.00	0.10	0.26	0.00	
HTC 4	S1(1)_pt3	68.47	24.85	0.00	2.13	1.34	2.39	0.18	0.03	0.12	0.42	0.06	
HTC 4	S1(1)_pt4	72.46	23.58	0.00	0.19	1.71	0.44	0.27	0.17	0.45	0.73	0.00	
HTC 4	S1(1)_pt5	68.44	24.38	0.17	0.44	3.27	0.78	0.45	0.13	0.58	1.27	0.10	
HTC 4	S1(2)_pt1	0.00	63.57	5.58	0.00	24.30	0.10	0.00	0.00	0.26	6.19	0.00	
HTC 4	S1(2)_pt2	67.15	27.39	0.00	0.26	2.61	0.60	0.30	0.08	0.37	1.25	0.00	
HTC 4	S1(2)_pt3	65.39	26.89	0.00	0.55	3.58	0.91	0.39	0.07	0.41	1.69	0.12	
HTC 4	S1(2)_pt4	68.72	25.20	0.08	0.47	2.80	0.77	0.39	0.13	0.38	0.98	0.08	
HTC 9	S2(1)_pt1	57.48	30.38	0.00	0.15	11.67	0.14	0.00	0.00	0.00	0.17	0.00	
HTC 9	S2(1)_pt2	47.46	28.23	0.00	0.19	0.26	7.34	0.12	0.43	0.00	14.78	0.14	
HTC 9	S2(1)_pt3	63.90	23.22	0.00	0.06	11.54	0.24	0.11	0.00	0.09	0.72	0.12	
HTC 9	S2(1)_pt4	74.75	21.83	0.00	0.30	1.08	0.26	0.80	0.00	0.08	0.65	0.26	
HTC 9	S2(1)_pt5	71.41	25.71	0.00	0.16	1.39	0.23	0.27	0.00	0.04	0.67	0.12	
HTC 9	S2(1)_pt6	72.34	25.63	0.00	0.13	0.66	0.13	0.39	0.00	0.03	0.53	0.15	
HTC 9	S2(2)_pt1	73.83	21.73	0.00	0.15	2.61	0.32	0.49	0.00	0.14	0.60	0.12	
HTC 9	S2(2)_pt2	66.16	22.26	0.00	0.12	10.85	0.19	0.00	0.00	0.03	0.34	0.06	
HTC 9	S2(2)_pt3	56.22	28.94	0.00	0.06	14.37	0.12	0.00	0.00	0.03	0.20	0.06	
HTC 9	S2(2)_pt4	33.73	46.72	0.00	9.72	0.12	9.23	0.00	0.00	0.31	0.17	0.00	
RD 7	S3(1)_pt1	46.15	38.64	0.00	5.18	9.13	0.16	0.00	0.05	0.18	0.31	0.21	
RD 7	S3(1)_pt2	76.08	15.37	0.00	0.33	3.89	0.88	0.76	0.11	0.59	1.98	0.00	
RD 7	S3(1)_pt3	61.14	21.85	0.05	0.41	13.38	0.52	0.30	0.14	0.84	1.36	0.00	
RD 7	S3(1)_pt4	59.64	31.44	0.00	1.22	1.89	1.49	0.58	0.10	0.69	2.80	0.16	
RD 9	S4(1)_pt1	32.60	46.21	0.00	9.66	0.10	10.93	0.00	0.00	0.24	0.16	0.00	
RD 9	S4(1)_pt2	34.85	43.10	0.00	0.11	21.03	0.18	0.00	0.00	0.31	0.42	0.00	
RD 9	S4(1)_pt3	68.24	28.11	0.00	0.39	0.76	0.82	0.49	0.00	0.34	0.78	0.08	
RD 9	S4(1)_pt4	50.87	35.45	0.00	5.93	0.54	6.71	0.00	0.00	0.10	0.29	0.05	
RD 9	S4(2)_pt1	36.96	9.97	0.00	0.07	1.19	0.19	0.29	0.00	0.12	0.32	35.22	metal particle
RD 9	S4(2)_pt2	36.50	44.48	0.00	4.36	1.02	0.29	0.14	0.00	0.14	12.85	0.16	
RD 9	S4(2)_pt3	69.35	27.56	0.00	0.24	1.05	0.41	0.48	0.00	0.36	0.43	0.12	
RD 9	S4(2)_pt4	0.00	50.72	0.00	16.79	1.21	27.83	0.26	0.00	0.97	1.49	0.41	

Table S5 Order of analysis of the liquid samples in the continuous TOC/TN autosampler run, with elapsed time since the first sample. HTC sample numbers refer to CCD runs. The last nine samples (Runs 22–30, from 9.1 h onwards) returned systematically low values (manuscript Section 3.5)

Order	Sample	Elapsed (h)	TOC (mg/L)	TN (mg/L)
1	RD1	0.00	4420	1393
2	RD2	0.32	4248	1286
3	RD3	0.63	3647	1216
4	RD4	0.96	4890	1392
5	RD5	1.37	7504	2236
6	RD6	1.69	10458	2715
7	RD8	2.02	8429	2116
8	RD9	2.30	10398	2661
9	RD10	2.56	7779	2206
10	HTC1	2.88	7937	2252
11	HTC2	3.21	7846	2273
12	HTC3	3.57	7516	2193
13	HTC4	3.85	11007	3045
14	HTC5	4.12	2058	820
15	HTC6	4.43	4088	1300
16	HTC7	4.78	4649	1351
17	HTC8	5.10	2963	1157
18	HTC9	5.41	4469	1270
19	HTC10	5.83	7866	2443
20	HTC11	6.15	10404	2710
21	HTC12	6.43	10063	2627
22	HTC13	6.75	7855	2244
23	HTC14	7.02	3335	1364
24	HTC15	7.30	7722	2223
25	HTC16	7.67	7571	2076
26	HTC17	8.00	11417	3167
27	HTC18	8.26	1802	792
28	HTC19	8.59	7471	2156
29	HTC20	8.87	7523	2141
30	HTC22	9.13	878	930
31	HTC23	9.48	927	952
32	HTC24	9.78	956	942
33	HTC25	10.18	885	826
34	HTC26	10.54	791	826
35	HTC27	10.83	673	651
36	HTC28	11.17	466	584
37	HTC29	11.53	362	521
38	HTC30	11.94	929	1216

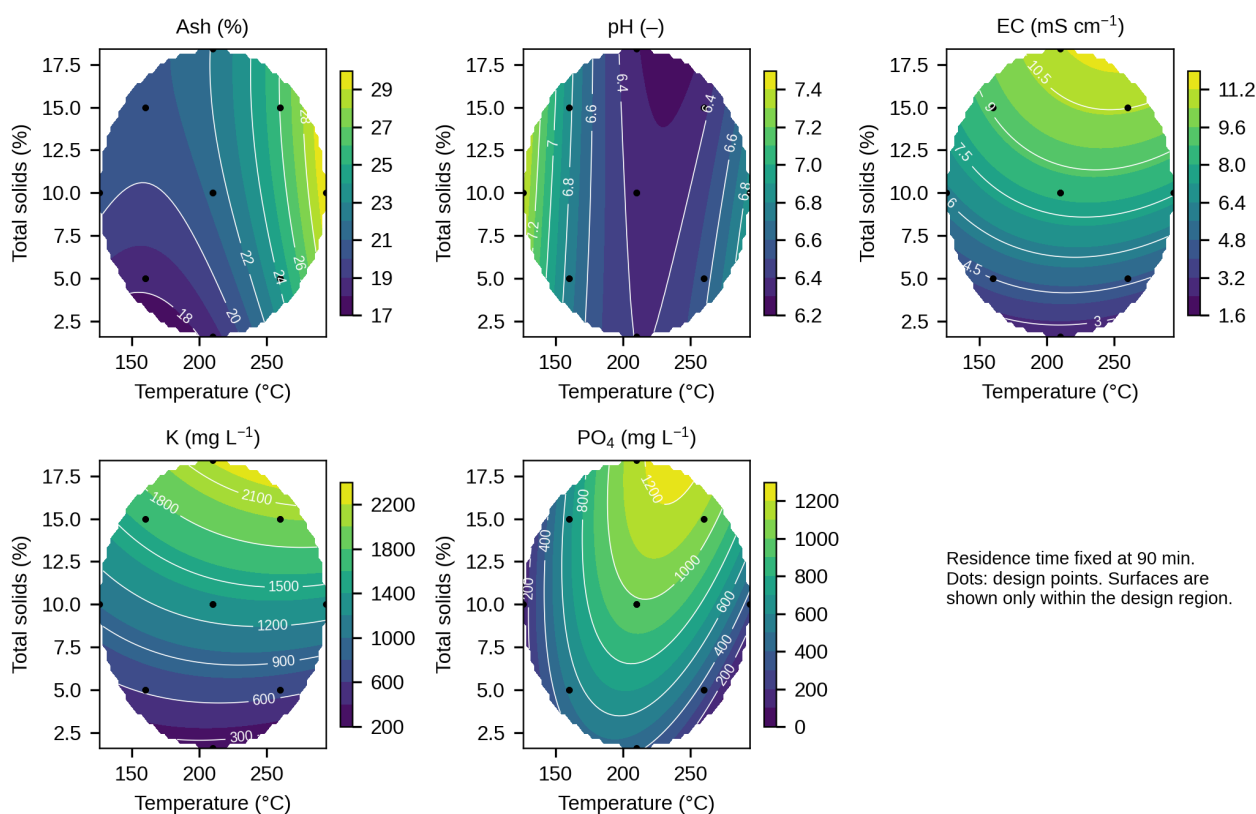


Fig. S1 Contour plots of the fitted response surface models for ash, process-water pH, electrical conductivity, dissolved K and dissolved PO₄ as functions of HTC temperature and total solids at a residence time of 90 min. Dots mark the design points

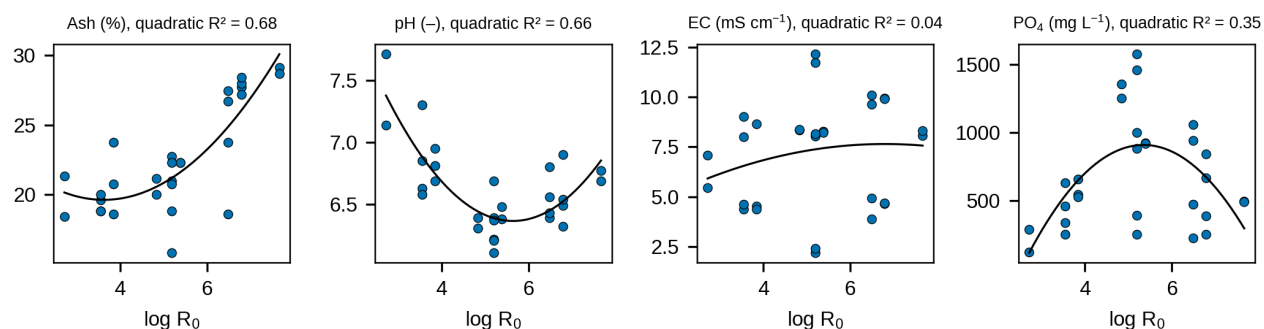
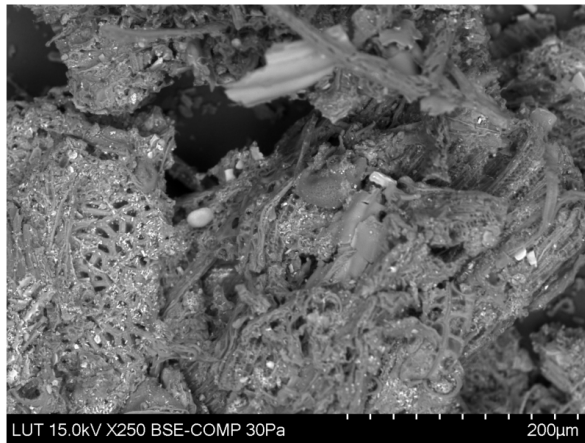
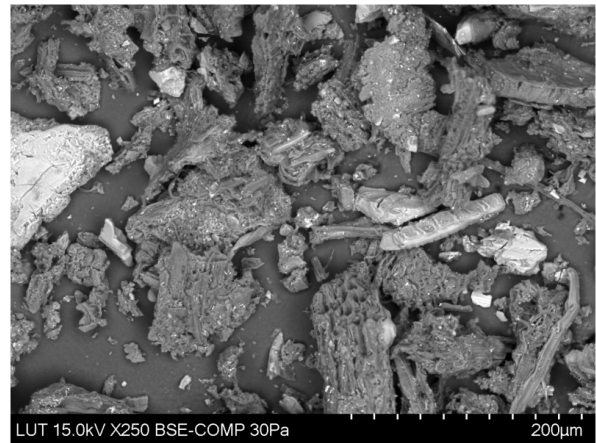


Fig. S2 Ash, process-water pH, electrical conductivity and dissolved PO₄ of the HTC products as functions of the severity factor log R₀, with quadratic fits

(a) RD 7



(b) RD 9



(c) HTC 4 (125.9 °C)



(d) HTC 9 (210 °C)

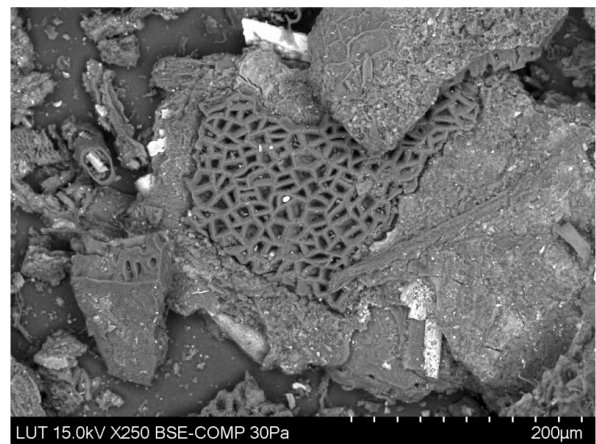
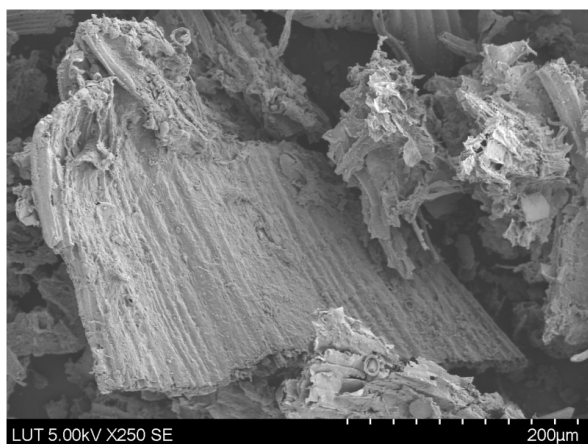
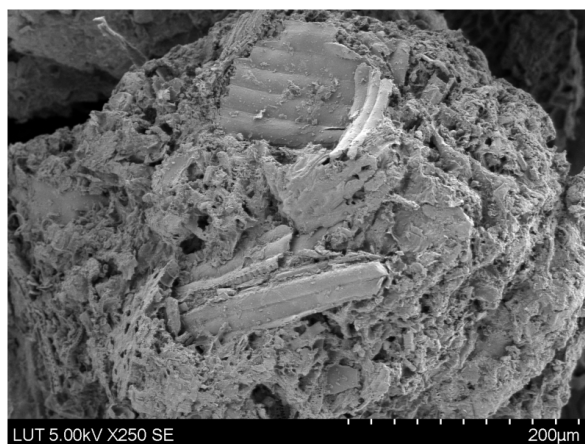


Fig. S3 Backscattered-electron micrographs (BSE-COMP, 15 kV, 30 Pa, $\times 250$) of RD 7 (a), RD 9 (b), HTC 4 (c) and HTC 9 (d). Brighter regions indicate higher mean atomic number (mineral inclusions)

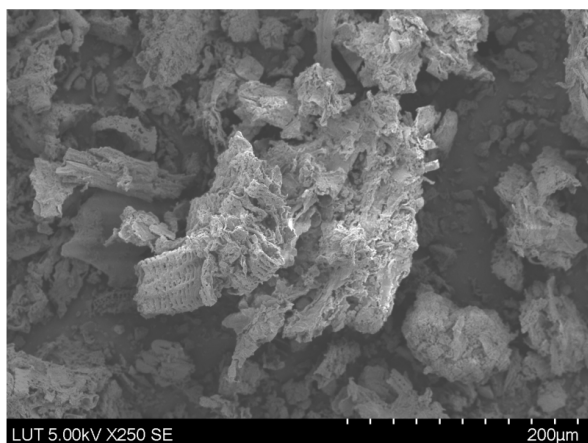
(a) RD 7



(b) RD 9



(c) HTC 4 (125.9 °C)



(d) HTC 9 (210 °C)

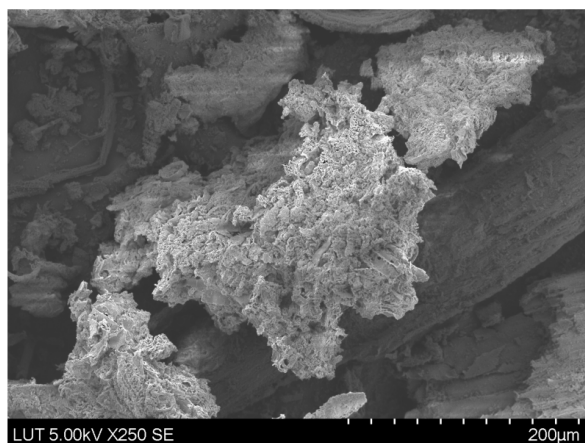


Fig. S4 Secondary-electron micrographs (5 kV, $\times 250$) of RD 7 (a), RD 9 (b), HTC 4 (c) and HTC 9 (d)

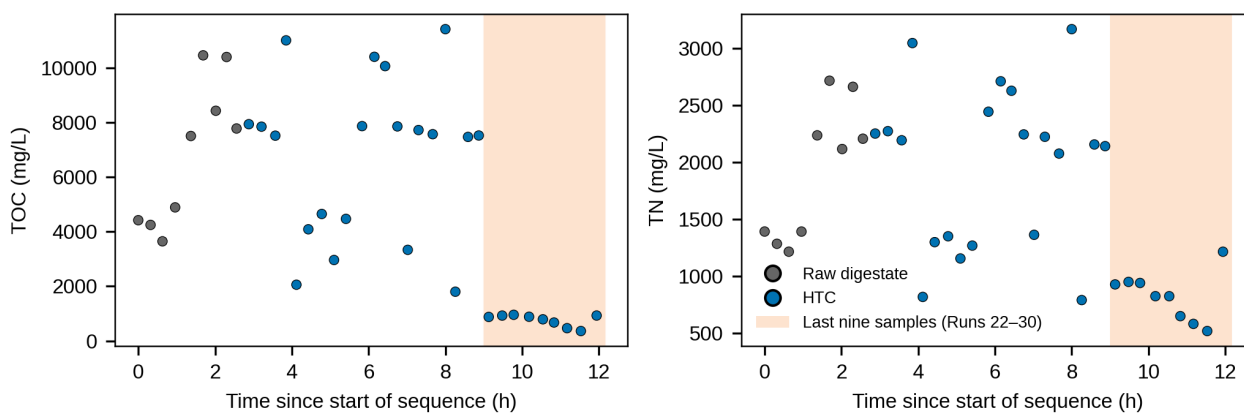


Fig. S5 Dissolved TOC and TN as a function of time since the start of the autosampler sequence. The shaded region marks the last nine samples (Runs 22–30)

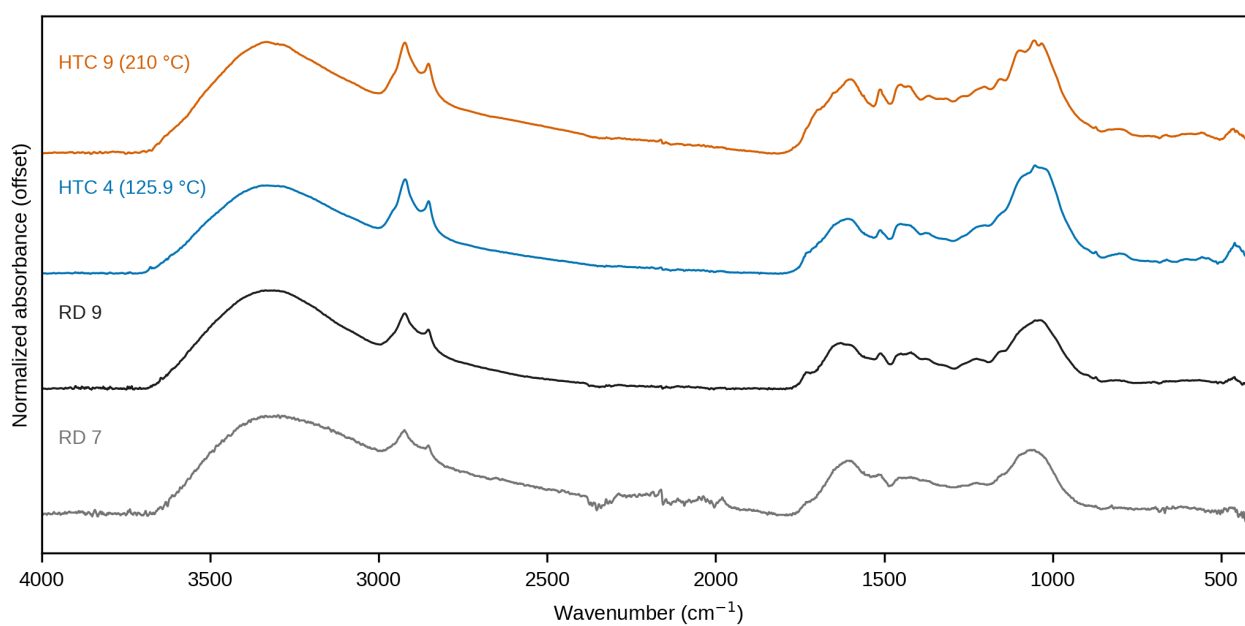


Fig. S6 Full-range normalized FTIR absorbance spectra (4000–400 cm⁻¹) of the four characterized cakes, vertically offset for clarity